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THIRD SEMESTER B.Tech EXAMINATION, NOVEMBER-2024

APPLICATIONS OF AI

Time: 3 Hours **Maximum Marks: 100**

Instructions: i. Missing data may be suitably assumed.

ANSWER ALL QUESTIONS

		PART - A			**5x2=10**
1.	**a.**	Define Data analytics.	L1	CO1	**02**
	b.	Define AI powered drones.	L1	CO2	**02**
	c.	List the key benefits of using AI in diagnostics.	L1	CO3	**02**
	d.	List the benefits of AI in food Processing.	L1	CO4	**02**
	e.	List the benefits provided by AI in customer service.	L1	CO5	**02**
		PART - B			**6x5=30**
2.	**a.**	Distinguish between Programming with AI and without AI.	L2	CO1	**05**
		OR			
	b.	Explain the different types of intelligence.	L2	CO1	**05**
3.	**a.**	Discuss the different applications of Precision farming.	L2	CO2	**05**
		OR			
	b.	Describe the key components of sensor-based agriculture.	L2	CO2	**05**
4.	**a.**	What role does AI play in drug discovery and development.	L2	CO3	**05**
		OR			
	b.	How does medical data management ensure the effective administration and governance of healthcare data?	L2	CO3	**05**
5.	**a.**	Differentiate between electric noses and next generation sequencing	L2	CO4	**05**
		OR			
	b.	Define AI based online Restaurant search engine and its features.	L2	CO4	**05**
6.	**a.**	Describe the role of AI in fraud detection.	L2	CO5	**05**
		OR			
	b.	Describe the different techniques used in recommendation engines.	L2	CO5	**05**
7.	**a.**	What are the key stages involved in medical data processing	L2	CO3	**05**
		OR			
	b.	Explain the main challenges of implementing AI in healthcare.	L2	CO3	**05**
		PART - C			**6x10=60**
8.	**a.**	Explain the components of machine learning and deep learning.	L2	CO1	**10**
		OR			
	b.	Explain the steps used by AI for data analysis.	L2	CO1	**10**
9.	**a.**	Discuss the different research opportunities of AI in agriculture.	L2	CO2	**10**

		OR			
	b.	Discuss the various applications of AI in crop monitoring and disease detection.	L2	CO2	**10**
10.	**a.**	How is hospital management and operations being enhanced by AI?	L2	CO3	**10**
		OR			
	b.	How is AI transforming medical imaging, and what are the key benefits with its implementation in healthcare diagnostics.	L2	CO3	**10**
11.	**a.**	Explain how AI is used for demand forecasting.	L2	CO4	**10**
		OR			
	b.	Explain the role of AI in food safety.	L2	CO4	**10**
12.	**a.**	Explain the tools and techniques used for AI fraud detection.	L2	CO5	**10**
		OR			
	b.	Explain the different ways in which AI can be used in sales and marketing	L2	CO5	**10**
13.	**a.**	Explain the key components of AI ecosystem.	L2	CO2	**10**
		OR			
	b.	Differentiate between conventional and precision agriculture	L2	CO2	**10**